Individual Research Log Book

Decentralized Multi Robot Collaboration

Course	CMPE 295A
Project	Decentralized Multi Robot Collaboration
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Submission date	10/28/2026

Project Overview

Our project studies whether two robots can work together without directly sending messages to each other. The project uses a vision-language-action model. The model receives camera images, the robot state, and a task instruction, then produces robot actions.

The first stage is a controlled single-robot baseline. The team is using a simulated SO-101 arm with SmolVLA, LeRobot, Isaac Sim 4.5.0, and Isaac Lab. This baseline must be understood before the same policy is later placed on two robots.

My individual contribution is the evaluation-and-data lane. I own the dataset audit fields the scorer needs, the success metrics, the evaluation wrapper and per-episode CSV schema, the B0/B1 run plans, the failure-label sheet, and the Workbook 2 chapters tied to that lane (evaluation methodology, implementation progress, and schedule). I am not claiming ownership of the Isaac Lab stand-up, the SmolVLA policy-server path, or the literature stack. Those belong to Ritwik, Snehil, and Bharat.

Current Status

Completed so far	Dataset audit, metric definitions, evaluation wrapper skeleton, B0/B1 command plans, failure-label categories, Workbook 2 evaluation and schedule chapters, and this individual research log.
Not completed yet	Scored B0/B1 rollouts on the A40 pod, horizon sweep, fine-tuned B3 score, and labeled failure videos from real episodes.
Important limitation	No final success rate or model comparison is claimed in this log because the closed-loop experiments are still pending on the simulator host.

Project Tools and Materials

- Git repository and project documentation, including Lab_log.md, team_workflow.md, and phase1/.
- NVIDIA Isaac Sim 4.5.0 and Isaac Lab for simulation (Phase 1 host: RunPod A40).
- LeRobot and SmolVLA for the vision-language-action policy (laptop checks used LeRobot 0.6.1).
- SO-101 simulated robot and the LightwheelAI/leisaac-pick-orange dataset (60 episodes).
- HPC3 H100/A100 queues for planned training; RT-core GPU for simulation and evaluation.
- Evaluation wrapper sitting on LeIsaac policy_inference.py and LeRobot async policy server.

Weekly Research Entries

The entries below describe the work completed on my lane. Dates roughly follow the Workbook 1 window through the Workbook 2 draft (late September to October 28).

Week 1 Understanding the Project Goal and My Lane

Activity	Reviewed the project description, team presentation, advisor notes, and lane assignment
Status	Completed

I studied the main project question: whether two robots can cooperate without direct robot-to-robot communication. I learned that the later multi-robot experiment depends on first building a reliable single-robot baseline with a measured success rate.

My contribution this week was to lock what “evaluation and data” means for me. I own the numbers and the records. I do not own the Isaac install or the policy-server wiring. That split kept me from waiting on the pod for work I could finish on a laptop.

Evidence reviewed: team reference presentation, project abstract, and team meeting notes. Result: I understood the project objective and where my lane starts.

Week 2 Pulling Fixed Evaluation Parameters

Activity	Extracted every parameter that must stay fixed before a score means anything
Status	Completed

I went through the workbook evaluation chapter and put the fixed knobs on one sheet: episode length 120 s, step rate 30 Hz, both cameras on, language instruction held constant, at least three pooled runs of five rounds, seeds recorded and reused, action horizon published per checkpoint, and simulator / LeRobot versions written down.

I also wrote that success has to come from the environment’s own check through the LeIsaac-SO101-PickOrange-Direct-v0 variant, not a distance rule we invent later when the built-in check is inconvenient.

My contribution was the run-metadata sheet the team can reuse without reopening the chapter. Evidence reviewed: Workbook 1 Chapter 6 and team_workflow.md. Result: the scoring controls were listed before any episode ran.

Week 3 Auditing the Dataset

Activity	Examined LightwheelAI/leisaac-pick-orange for evaluation-relevant fields
Status	Completed

I audited the dataset the baseline is scored on. It has 60 episodes and about 36,293 frames at 30 fps. Robot type is so101_follower. Front and wrist cameras are 480×640. Action and state are six floats in this order: shoulder pan, shoulder lift, elbow flex, wrist flex, wrist roll, gripper. Median episode length is about 18 s. Three oranges per episode. The task string in the dataset is “Grab orange and place into plate.”

One open question I raised with Ritwik: the demos end with a return to rest. If -Direct-v0 only tests placement and ignores that return, our numbers and the published ACT numbers are not measuring the same thing. That check is still open.

My contribution was connecting the dataset fields to the later CSV columns so the evaluator records which camera contract, instruction, and action order produced each result. Evidence reviewed: Hugging Face dataset card, episode samples, and phase1/01_dataset_exploration notes. Result: evaluation-relevant interface details recorded.

Week 4 Defining the Metrics

Activity	Wrote the headline metrics and the reporting rules for every baseline row
Status	Completed

I defined per-orange success as oranges placed divided by three times the number of episodes. I also require the per-episode count from 0 to 3 as a histogram, because the distribution on this task is bimodal and a mean alone hides that. A 95 percent confidence interval is reported with the numerator and denominator shown.

The planned rows are B0 random action, B1 zero-shot lerobot/smolvla_base, B2 external ACT numbers quoted only (44.0% per-orange on Isaac Sim 5.1, not rerun), and B3 fine-tuned SmolVLA on the same seeds and same LeRobot version as B1.

My contribution was making the comparison rules boring and hard to cheat by accident. Evidence reviewed: workbook evaluation chapter and published same-task ACT tables. Result: the metric draft the wrapper prints against.

Week 5 Building the Evaluation Wrapper Skeleton

Activity	Designed and drafted the wrapper over policy_inference.py and the async policy server
Status	Completed; closed-loop runs pending

I wrote the first version of the evaluation wrapper. It sits on top of LeIsaac's scripts/evaluation/policy_inference.py and talks to LeRobot's async policy server. Inputs are checkpoint ID, policy type, seeds, rounds, episode length, step rate, action horizon, and device.

Each episode writes one CSV row: run ID, seed, round, checkpoint, horizon, oranges placed, termination reason, simulator version, LeRobot version, and video path. I did not score a live episode this week. The contract test and the LeRobot pod pin were still open, so any number I printed would have been uninterpretable.

My contribution was the logging path itself, not a fake score. Evidence produced: wrapper skeleton and results_template.csv under phase1/05_evaluation/. Result: when the first rollout passes, scoring can start the same day.

Week 6 Run Plans, Horizon Rule, and Failure Labels

Activity	Finalized B0/B1 command plans, horizon sweep rule, and failure-label categories
Status	Completed; experiments pending

I wrote the B0 and B1 run plans so the commands exist before anyone is under time pressure on the pod. First rollouts use action horizon 16. The later sweep is 1 to 50, because smolvla_base returns at most 50 future actions per query. The chosen value is published with the score.

I also started the failure labeling sheet. Categories are collision, missed grasp, dropped orange, wrong placement, timeout, camera or action mismatch, and policy server disconnect. No labels yet, because nothing has been scored.

Camera mapping locked with the team notes: front → camera1, wrist → camera2, third SmolVLA camera key left unused. Instruction locked to the dataset string. Evidence reviewed: Lab_log.md decisions from 2026-10-26 and my scoring sheet. Result: the protocol is ready; only the pod episode is missing.

Week 7 Workbook 2 Pack and Personal Contribution Boundary

Activity	Prepared Workbook 2 evaluation / progress / schedule chapters and this research log
Status	Completed; implementation of scored runs pending

I selected Evaluation Protocol, Experiment Logging, and Baseline Scoring as my individual area. That is separate from building the full training pipeline or the future two-robot system. I drafted Workbook 2 Chapters 6–9 for the group merge, with bolded updates to the implementation plan and a Gantt/PERT view for both semesters, and pushed them on branch feature/kanaka_workbook2.

Context that landed on the team this week, not on my keyboard: the single-frame contract test passed on 2026-10-26. SmolVLA loads at 450.0M and returns six numbers in SO-101 order, chunk (1, 50, 6). That unblocks my trust in a future B1 score. It does not mean I have scored B1 yet.

Evidence produced: Kanaka_Workbook2_Contribution.pdf/.docx, phase1/05_evaluation/, and this log. Result: my personal contribution is written down and separated from simulator stand-up and model-server work.

Summary of Personal Contribution

My work so far has focused on making the planned baseline evaluation measurable and repeatable. I audited the dataset, locked the metrics, drafted the wrapper and CSV schema, wrote the B0/B1 plans, set the failure labels, and put the evaluation and schedule chapters into the Workbook 2 pack for the group.

The main value of this contribution is that later results can be compared fairly. A model score is not useful if the camera setup, action order, instruction string, reset process, simulator version, or success rule changes between runs. The log format and test plan are intended to preserve that information.

I have not claimed completed training or final performance results. The current status is research, design, and scoring preparation. The first closed-loop B0/B1 runs, the horizon sweep, the fine-tune score, and the labeled failure videos remain future work on my lane once the A40 rollout passes.

Pending Work

- Confirm with Ritwik that the -Direct-v0 success condition matches the three-orange goal, including return to rest.
- Write the final LeRobot version from the pod into Lab_log.md and use that same pin for B1 and B3.
- Run B0 (random action) and B1 (zero-shot SmolVLA) under the fixed protocol after the first rollout.
- Sweep action horizon (1–50), publish the chosen value, and keep seeds identical across checkpoints.
- Score B3 when the HPC fine-tuned checkpoint and norm stats arrive on the sim host.
- Label failure videos with the prepared categories and update this log with dates, commands, and evidence.

Evidence and References

- Project team reference presentation, project abstract, and final_project proposal notes.
- Project Git repository: branch feature/kanaka_workbook2 and phase1/05_evaluation/.
- Lab_log.md and team_workflow.md (including 2026-10-26 contract-test decisions).
- LightwheelAI/leisaac-pick-orange dataset information and phase1 dataset report.
- Hugging Face LeRobot and SmolVLA documentation; lerobot/smolvla_base checkpoint.
- NVIDIA Isaac Lab / LeIsaac evaluation path (policy_inference.py).
- Workbook 1 group submission and Kanaka Workbook 2 contribution pack.
- Prior individual log covering Sep 29–Oct 26 evaluation-lane hours.

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